

Challenge archive: SekaiCTF 2022 - FaILProof

deuterium — Event date: 2022-09-30

Challenge event began 2022-09-30; article publication date not recorded.

Canonical HTML: <https://deut-erium.github.io/challenges/sekaictf-2022/failproof/>

Abstract

SekaiCTF 2022 - Cryptography.

By deuterium.

Original description

I have designed a failproof encryption system with possibly arbitrarily small public keys. I will be as famous as Et Al one day, but only if I can somehow figure out a decryption mechanism...

Read this challenge as [plain text](#) or [JSON](#).

Files

- [source.py](#) ([archived server source](#)).
- [Fixed instance](#) ([output.txt](#)).

The flag module is needed only to generate a new instance. Solve from the source and the two lines in output.txt; no server is needed.

This fixed output was generated for the post with the unchanged original program and event flag. It is not a recording from the competition.

Download

```
mkdir -p sekaictf-2022-failproof
cd sekaictf-2022-failproof
curl --fail --location --output challenge.txt \
  "https://deut-erium.github.io/challenges/sekaictf-2022/failproof/challenge.txt"
curl --fail --location --output challenge.json \
  "https://deut-erium.github.io/challenges/sekaictf-2022/failproof/challenge.json"
curl --fail --location --output source.py \
  "https://deut-erium.github.io/assets/challenges/sekaictf-2022-failproof/source.py"
curl --fail --location --output output.txt \
  "https://deut-erium.github.io/assets/challenges/sekaictf-2022-failproof/output.txt"
```

Check your flag

Enter the flag

The local SHA-256 checker requires JavaScript. The challenge itself does not.

Sources and solutions (spoilers)

- [Full archive](#) (may contain flags).
- [Solution](#)
- [Original description or source](#)
- [Event credits](#)
- [CC BY-NC-SA 4.0 \(description\)](#); [AGPL-3.0-or-later \(source\)](#).
- [Local AGPL-3.0 license](#)
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[All challenges](#) · [CTF tutorials](#)

| Interactive controls are omitted from this PDF. See the canonical HTML version.